

System architecture

AI-powered equity research platform. One Next.js monolith, two entry paths, on a self-managed VPS.

CLIENT / SCHEDULING

Browser
Submits a ticker, receives the rendered report.

GitHub Actions
Weekday *cron* + manual *workflow_dispatch*. Bearer-secret HTTPS call.

EXTERNAL SERVICES

DNS

Cloudflare DNS A record *analysts.korestandard.com* → VPS IP. DNS-only (grey cloud), not proxied.

INFRASTRUCTURE

SELF-MANAGED VPS · DOKPLOY (DOCKER PAAS)

Traefik — managed by Dokploy
Let's Encrypt TLS termination. Routes :443 → app container :3000.

Next.js monolith — App Router + Route Handlers

ONE CONTAINER · NO SEPARATE BACKEND

1 · TICKER RESEARCH ASSISTANT on demand

app/research/[ticker]/page.tsx
Server component. Renders the two-block report.

api/research/[ticker]/route.ts
Cache lookup → Finnhub → Groq → cache write.

lib/report-schema.ts
Zod-validates model JSON; splits facts from narrative.

2 · DAILY STOCK IDEA ENGINE scheduled

api/screen/route.ts
Bearer-secret guarded. Orchestrates the run.

lib/universe.ts
Fixed ~50-ticker list, fetched sequentially.

lib/score.ts NO LLM
Deterministic ranking in plain TypeScript.

lib/db Drizzle ORM — typed queries and migrations, shared by both paths.

Postgres
Drizzle-managed schema on a Docker volume. *reports_cache* · *daily_ideas* · *screen_runs*.

Environment variables
API keys, *DATABASE_URL*, cron secret. Injected by Dokploy; never committed.

SQL over the Docker network

Finnhub
Quotes, financials, company news. The only source of numbers in the product.

Groq
Inference only. *llama-3.3-70b-versatile* for the report, *llama-3.1-8b-instant* for the daily thesis. Prose, never ranking.

Resend
One transactional email per weekday run.

Reached outbound over HTTPS from the app container. No inbound third-party access to the host.

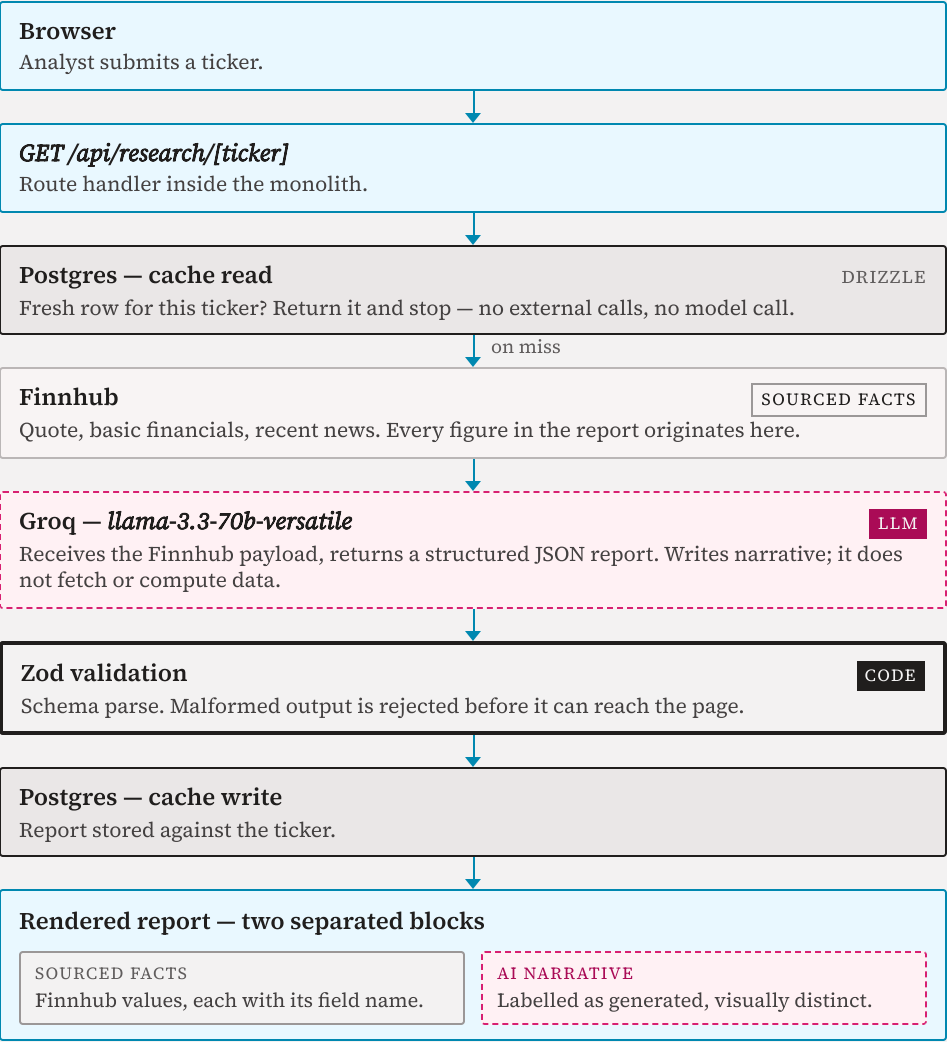
Two request paths

Read down each column. Where a model is called — and where one deliberately is not.

-  Deterministic code — same input, same output
-  LLM call — non-deterministic, validated on return
-  Third-party API
-  Datastore

1 Ticker Research Assistant

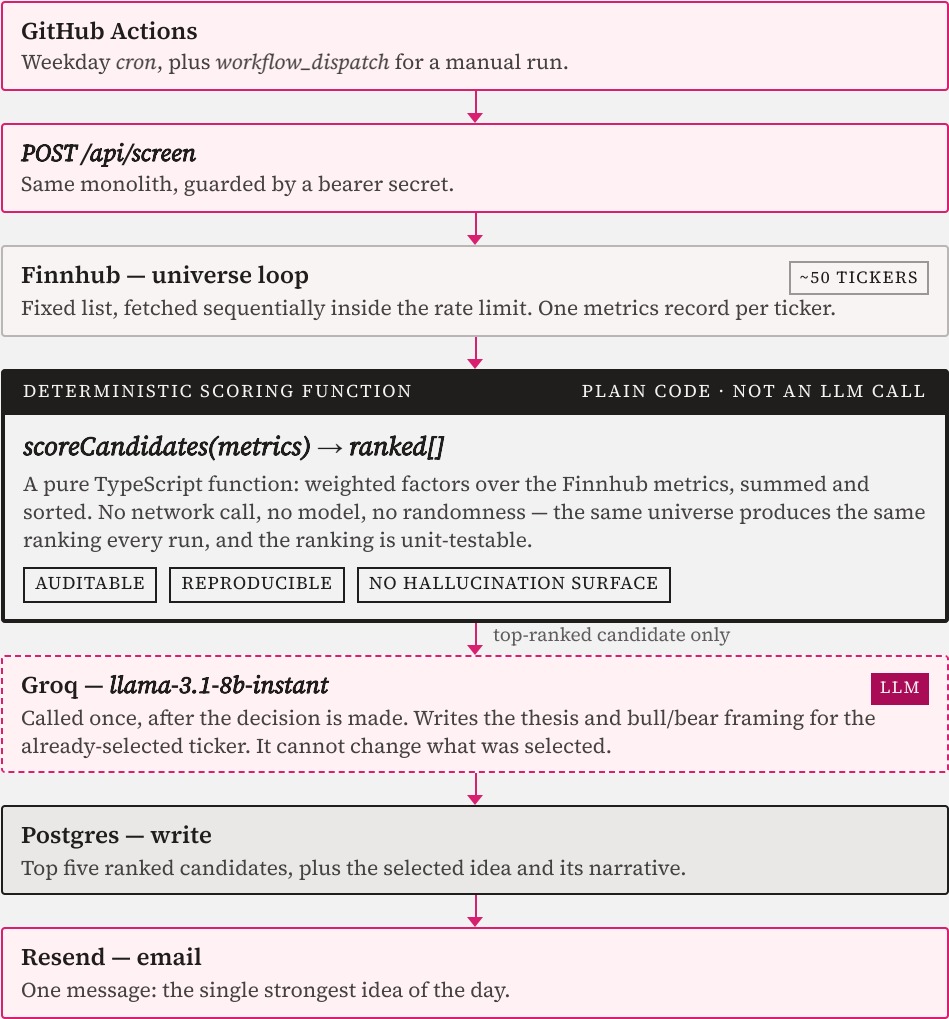
User-triggered · synchronous · one ticker



Path 1 — what the model can affect. Wording only. Numbers are echoed from Finnhub into their own block; the narrative sits in a second, labelled block. A schema failure drops the narrative rather than the page.

2 Daily Stock Idea Engine

Scheduled · batch · ~50-ticker universe



Path 2 — what the model can affect. Wording only. Selection and ranking complete before any model is called, so a hallucination can misdescribe the chosen idea but can never change which idea was chosen.